

UDAY SHANKAR ROY

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Research Interest

- Machine Learning
- Trustworthy AI
- ML for Security
- Computer Vision

Research Experience

- ***“Deep Learning Approaches for Skin Lesion Hair Segmentation: Addressing Diversity through Dataset Adaptation to Brown Skin Tones”***
Uday Shankar Roy, Sk Imran Hossain, Md. Abdus Salim (IEEE QPAIN 2026)
 - For brown skin, there is no available skin lesion hair dataset, which can degrade results on brown skin images. The available dataset was augmented to improve performance on brown skin tones.
 - Evaluated U-Net, Attention U-Net, Nested U-Net, LinkNet, SegNet, and ENet for skin lesion hair segmentation.
- ***“Frequency-Domain AI-Generated Image Detection: Exploring Decoder and Channel Attention for Feature Refinement”***
Uday Shankar Roy, Mahbuba Jahan Minu (IEEE OMLET 2026)
 - Explored a U-Net and Attention U-Net decoder with an EfficientNet-B0 encoder for AI-generated image detection.
 - Investigated whether channel attention can improve AI-generated image detection in the frequency domain.
- ***“Unmasking Shortcut Learning in IoT Intrusion Detection: A Forensic, Multi-Paradigm Evaluation of Feature Dependence and Data Leakage”***
Uday Shankar Roy, Mahbuba Jahan Minu (Under Review)
 - Showed that ML-NIDS models can achieve near-perfect performance by exploiting dataset shortcuts (like static IP addresses and timestamps) instead of actually learning attack patterns.
 - Found that different types of models rely on these shortcuts to very different degrees. For example, tree-based models could exploit timestamp patterns, while linear models could not.
- ***“PromptTriage: A Multi-Class Detection and Defense Framework for Prompt Injection Attacks in LLM-Integrated Applications”***
Mahbuba Jahan Minu, Uday Shankar Roy (Under Review)
 - Developed a multi-class framework that categorizes prompt injection attacks into four classes and maps each class to a type-specific defense action.
 - Evaluated the framework against a Qwen2.5-1.5B-Instruct model, reducing attack success from 16/50 to 1/50 while blocking or sanitizing 48 of 50 attacks.

Academics

Khulna University of Engineering & Technology 2019-2024
B.Sc. in Computer Science and Engineering

- Institutional CGPA: 3.48 out of 4.0 (**3.75** in last two semesters)
- FCSA Evaluation (U.S. Equivalency):
Overall GPA: 3.82 out of 4.0
Upper Division GPA: 3.96 out of 4.0

Khulna Public College 2015-2018
Higher Secondary Certificate in Science
GPA: 5.00 out of 5.00

Professional Experience

System Engineer July 2024-Present
E-Zone Limited

- Manage 2600+ carrier-grade datacom nodes for Robi Axiata Bangladesh Ltd. across the country

- Prepare operational MOPs for live network changes
- Handle service requests and ensure timely resolution
- Troubleshoot issues and provide detailed root cause analysis (RCA)

Standardized Test Score

International English Language Testing System (IELTS) – 29 June, 2025

Overall	Listening	Reading	Writing	Speaking
7.5	7.5	7.5	7	7

Technical Skills

- **Programming Languages:** Python, C, C++
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn
- **Data Analysis & Computer Vision:** NumPy, Pandas, Matplotlib, OpenCV
- **Web Based:** ASP.NET, JavaScript, HTML, CSS
- **DBMS:** SQL, MySQL
- **Systems & Tools:** Linux, Windows, Jupyter Notebook, Google Colab, Git
- **Others:** OpenGL, LaTeX

Projects

- **NewsPro- Autonomous News Aggregation Platform**
Technologies: Python, FastAPI, Google Gemini API, SQLite, Turso Cloud, BeautifulSoup, Tailwind CSS
An automated news platform that continuously ingests breaking stories from major news portals, clusters duplicate coverage, synthesizes reports using Google Gemini and automatically publishes them.
- **Human Step Detection**
Technologies: Python, Scikit-learn, Numpy, Jupyter Notebook
Detects human steps from the data of phone's accelerometer and gyroscope.
- **KUET Shaheed Minar**
Technologies: OpenGL 3.3, C++
A visual representation of KUET Shaheed Minar using OpenGL which contains various dynamic objects.
- **QuizMate-Online Quiz System**
Technologies: ASP.NET, C#, Microsoft SQL Server
A web application for conducting online quiz.
- **Build or Break**
Technologies: Python, PyGame
Build or Break is a Python-based strategy game.

Co-curricular Activities

- Vice President at Software Research Development Community of KUET (Bit2Byte) 2023-2024
- Web Manager at Special Group Interested in Programming Contest (SGIPC), KUET 2023-2024
- Volunteer at Hult Prize at KUET 2019

Achievements

- **Dean's List Award, Fourth Academic Session, KUET** 2024
- **AWS AI & ML Scholarship** 2023
- 2nd Place, AWS Deepracer Student League 2023
- 13th Place, UGV South Programming Contest 2023
- 11th Place, Intra KUET Programming Contest 2022
- 2nd Place, EEE Day IT QUIZ, KUET 2019